Energy Management Scheme Using Prediction Algorithms in Energy Harvesting Wireless Sensor Networks

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Under the guidance of

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CERTIFICATE

This is to certify that the project entitled **"Energy Management Scheme Using Prediction Algorithms in Energy Harvesting Wireless Sensor Networks"** submitted by Aditya Sharma (B.Tech 2020) is based upon their work under the supervision of **Dr. Tarachand Amgoth** in the Department of Computer Science and Engineering, Indian Institute of Technology (Indian School of Mines) Dhanbad and that neither the project report nor any part of it has been submitted for any degree/diploma or any other academic award anywhere before.

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Acknowledgements

I, a student of B. Tech Final Year of Computer Science and Engineering, Indian Institute of Technology (Indian School of Mines), Dhanbad am glad to present a project titled **"Energy Management Scheme Using Prediction Algorithms in Energy Harvesting Wireless Sensor Networks".**

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Abstract

A Wireless Sensor Network (WSNs) consists of finite number of energy constrained sensor nodes. In recent time, application of sensor network has been increased in many places, such as environment system monitoring, border surveillance, vehicle tracking, Internet of Things (IoT) and health monitoring etc. One major issue of the sensor network is energy management, because sensor node deployed in any harsh area. The charging capacity of typical sensor node battery is finite. Moreover, the replacement or recharging of the sensor node battery is often impractical, therefore; it may create dis-connectivity in the network.

Therefore, one solution has been discovered to the recharge sensor node battery that is called energy harvesting technique. Generally, ambient resource energy is non controllable and their prediction is very necessary for initial energy management. In this paper, we have proposed modified Pro-energy prediction technique to the control unnecessary error in the harvesting system. This method works based on the previous day of the time slots to the current energy status. However, management of the initial energy is quicker in this method. The technique is showing better result and beats other method, just as EMWA, WCMA and Pro-energy. In the simulations parts, different kinds result parameters have shown in the sensor network.

Keywords: Wireless Sensor Network, Energy harvesting, Prediction Techniques, Energy management, Solar energy

Chapter 1: Introduction

The Wireless Sensor Networks (WSNs) includes a collection of low constrained sensor nodes and performing different task in the real time application. The application can be possible different kinds, such as home monitoring, border surveillance, environments system tracking, Internet of Things (IoT) and vehicle tracking etc [1]. The sensor node device is made different types of practical component, which is the transceiver unit, power unit, sensing unit and analog to digital converter (ADC). Out of the four components, power unit is very critical, because there are so many tasks performed between sensor node and the sink node using routing techniques [2].

The routing technique may be single-hop routing or multi-hop routing. During the routing process, the consuming of energy rate is very rapidly in the sensor network [3, 4]. The main sources of energy depletion in the sensor network have included overhearing, re-transmission of the packet, idle listing and control packet overhead. Moreover, battery of the traditional sensor node is non-rechargeable. It is powered by minimum capacity due to the short-range recharge cycle. Generally, sensor node is deployed in the unreachable place; thereby replacement of battery of node is too much difficult task [5]. Research has been solved the limitation of the energy resources that is called energy harvesting (EH) technique. The EH techniques can be different ways, such as solar, wind, mechanical and thermal [6]. However, it is very effective solution in the sensor network with the less cost. In order to handle the, just a bottleneck problem of the energy resources in all the limited energy system, then renewable energy system provides unlimited energy resources in the WSNs [7].

Besides, general principle of the harvesting system is to convert an ambient resource into the electricity, and is supplied to the energy to the sensor nodes [8]. In the EH system, there are so many components used, namely rechargeable battery, super capacitor, harvesting transducers, predictor and power management unit (PMU). The rechargeable battery can be different types, such as sealed lead acid (SLA), nickel metal hydride (NiMH), nickel cadmium, lithium polymer Li-polymer, and lithium-Ion-Li-Ion. Super-capacitor is also additional storage

class and sometimes itis referred to hybrid storage class, when its combined with together [9]. In the harvesting system, role of energy predictor is very important, because an ambient energy resource is uncountable and their prediction must for effective energy management in the WSNs. Moreover, solar harvesting system and wind harvesting system fully depends outdoor system, means solar energy is based on the sunny day and wind energy depends mass of air. Because, harvesting technique is very dynamic system, harvesting rate always changes according to the weather condition, therefore; using prediction technique, we can manage the uncontrolled energy.

The PMU component is key of the harvesting system, because its work just central point of the harvesting circuit. In the harvesting technique, there are two options to power supply for the sensor node, such as harvest spend and harvest store spend. The harvest spends process justified to use an energy resource, just at time, when needed. The harvest store spend technique works based on the future use of the energy resources. However, second option is very effective and useful in the EH-WSNs. This paper is based on the prediction technique, and it is initial method for energy management scheme. Generally, ambient energy is very uncontrolled as well as dynamic therefore, prediction process must to design efficient algorithm.

There are different parameters of ambient resources in the EH-WSN, such as uncontrolled but predictable, partially controllable, un-controllable and unpredictable and fully controllable. To identify the above problem of the harvesting system, role of prediction is very prime to store efficient energy. Moreover, prediction technique does not give exact result, but it will show approximate value, because ambient resources depends various factor of the weather conditions, just as intensity of the sun light, cloudy weather and high storm [10]. The practical use of the EH system justified to deal using the variable behaviour of the energy resources. It implies the amount of harvested energy and energy-rate over the diurnal time slot of the day. Therefore, prediction technique has been developed to adjust the power need for sensor device from ambient energy resources. Related to above issue problem, we have demonstrated a modified Pro-energy prediction technique to solve error problem in the EH-WSNs. This method works in the state to art manner in compare to existence techniques, just as EMWA, WMCA and Pro-energy.

Chapter 2: Literature Survey

There are many works related to prediction technique, which has done in the EH-WSNs. In [11], authors have proposed new model based on the solar harvesting system that is called Accurate Solar Irradiance Prediction Model. It works both term means, short term as well as long data term using real data sets. The proposed method is using non-linear model to show robust throughput in the sensor network. The ASIM mode is working with the help of Markov chain model and it is defined probability matrix, which has stored different state of the irradiation data. There are different performance metrics shown in this paper, such as energy storage level of the sensor node, rate of input and output energy and throughput. The authors have designed a solar powered wireless sensor network (SP-WSNs) to handle monitoring application [12].

It works basically in the hostile environment for self-power management and automation. There are two techniques used for efficient protocol, namely multi-hop routing SP-BCP and energy rate control scheme PEA-DLEX. One of famous prediction technique used in this paper that called Weather Condition Exponential Weighted Moving Average (WC-EWMA) and it is based on the solar harvesting system. Some other modelling is worked, such as off-line modelling and profiling, offline load function profiling and off-line battery modelling. The performance of method justified in the different types of the prediction error ratio for the various weight factor. They have demonstrated operational duty cycle to adjust power management in the sensor network [13]. This method is based on the solar EH system and each sensor node powered by a rechargeable battery with solar harvesting kit.

One well known method used in the proposed work that is called reinforcement learning to dynamic power management. This technique works based on the decision-making agent and three stages used to identify the final result, namely action, reward and state. Another state is used to justified EH-rate in the different type of weather conditions, such as low, medium and high. However, using this technique has been improved residual energy of the battery, thereby; effected network life time of the sensor nodes. In the [14], authors have introduced Pro-energy VLT to solve multi-sources EH prediction problem. This method works different types of data

set, such as wind data and solar data. Pro-energy VLT technique uses variable length time slots, because an ambient energy resource is dynamic system; therefore, used variable length day time slots. Two prediction modules have used in the proposed method, namely short-term energy prediction and medium-term energy prediction. Some other function worked in this paper, namely weight assignment, time-slot resizing and profile pool update. The performance of the algorithm gives better result for network energy management and remove bottleneck problem in the network.

The authors have introduced a novel type of solar energy prediction technique in the sensor network [15]. Here, it has proposed Q-learning based solar energy prediction (QL-SEP). The QL-SEP method works both condition prediction past days as well as current observation. However, proposed work is divided into the equal length slots and solar information kept in this slot. The performance of work is more effective in term of harvesting rate error. There are different error metrics measured in the work, such as prediction error ratio for various weight factor, prediction error for different slots and prediction error for the all months. They have demonstrated energy management with solar powered EH system in the WSNs [16]. Two algorithms used in the proposed work, namely Mixed Integer Linear Programming (MILP) and dynamic programming. Some other method used in the paper, such as forward dynamic programming technique and distributed greedy algorithm. The performance of the method has showed low computational power of memory.

Here, authors have introduced transmission technique using Hierarchical Least-Mean Square (HLMS) [17]. The proposed method has used to predict the sensed data in the sensor network, means sensor nodes to the base station. Some metric used to solve energy consuming rate, which are data prediction for error estimation, weight coefficient matrix and fractional calculus. Here, authors proposed neutral power management by using prediction technique and it is more effective in the sensor network [18]. One focus of the proposed algorithm is to manage energy budget of the sensor node. However, one famous technique used that is called Budget Assignment Principles (BAPs) to control Battery Residual Energy Level (BREL). Some other components used in the paper that are duty cycle, energy budget generation, slotted time system for energy budget and energy matching ability. In [19], authors have designed solar based prediction for effective energy management in the sensor network. One of the famous

prediction techniques used that is called Weather-Conditioned Moving Average (WCMA) and each sensor node powered by rechargeable capacitor. This method works based on the previous information of the weather condition as well as kept average solar information until present day. Some metrics have applied in the method, namely actuate/sample, process and send. The performance of work is better in the term of minimum error of EH-rate.

They have demonstrated a low power EH device to control the uncontrollable energy resources [20]. It is based on the long-term indoor system solar radiant energy measurement and here, taken battery energy level. There are two types energy storage, which are linear and nonlinear. Some important functions used in this paper for the effective EH system, such as Time Fair Rate Assignment (TFRA), Progressive Filling (PF) and Stochastic Energy Models. The performance of the technique improved to use of the rechargeable battery and super-capacitor. In [21], authors have proposed some previous prediction techniques, such as Exponential Weight Moving average (EWMA) and Weather Conditioned Moving Average (WCMA). In EMWA has showed previous day prediction energy with respect to different time slots of day. However, in this method calculated historical information of energy harvesting rate. In Eq. (1) has identified last amount of harvesting energy (H) and estimated energy (E) by the EMWA using weighting factor $0 < \alpha < 1$. If the value of α is greater, then less chance harvesting rate of energy and vice-versa. In Eq. *d* represent present day with *n* number of slots.

$$\mathrm{E(d,n)} = \alpha \mathrm{E(d-1,n)} + (1-\alpha)\mathrm{H(d-1,n)}$$

The different α calculated to show different seasonal changing of the weather condition. Moreover, the value of α is crucial in the prediction techniques. The WCMA is an extension of the EWMA prediction technique. In the WCMA method stores previous harvesting value using matrix E(i, j), with the ith day on the jth slot. However, WCMA method collects harvested data of the relevant slot of previous day. The value of α is a weight factor same as EMWA technique, then Eq. as follow:

$$\mathrm{E(d,n)} = \alpha \mathrm{H} + (1-\alpha)\mathrm{M}_D(d,n)GAP$$

Where, M_D has calculated mean value of sample of the previous day. The mean value of past day D as:

$$\mathrm{M(d,n)} = \frac{\sum_{i=d-D}^{d-1} E(i,n)}{D}$$

The main contribution of the WCMA is GAP_k value factor. In this method calculated to measure the solar harvesting rate condition of the present day to see previous day conditions. However, to the compute GAP_k factor, we first identify a vector V= [v_1, v_2, .., v_K] with the past K elements. Each sample harvested energy denotes the ratio of the harvesting energy to the average mean value.

$$V_k = \frac{E(d, n-K+k)}{M(d, n-K+k)}$$

However, it is more importance to the nearest values on the time. The WCMA technique has identified another vector P = [P_1, P_2, P_3, ... , P_K], where, each of vector defined according to distance to the actual slot.

$$P_k = \frac{k}{K}$$

Finally, the weight factor, GAP_K has computed:

$$GAP = \frac{V.P}{\sum P}$$

Chapter 3: Problem Statement

Section 3.1: Problem Formulation

In EH-system, one of the most challenging issues is energy prediction, because harvesting rate fully depends on the ambient weather conditions. The weather may be cloudy or sunny, wherein the sensor node maybe deployed in a harsh area. The solar harvesting technique and wind harvesting systems depends on the outdoor system, namely high solar light intensity and presence of speedy winds.

However, this technique is the initial process to control ambient resource management. If the ratio of prediction accuracy is sufficiently high, then residual energy of the battery will be high because energy predictor device directly connects with rechargeable battery of the solar harvesting device in the EH-WSNs. Generally, sensor node is scattered in the unreachable location and data transmitted to the base station in the regular time interval. Another possible issue may be, quality of the harvesting device, such as energy predictor, efficient rechargeable battery and super-capacitor.

Section 3.2: Solar Harvesting Prediction Model

In this section, we have described about the solar harvesting prediction model and all assumption is taken based on the proposed technique of this paper. Here, each sensor node is powered by a rechargeable battery with solar harvesting kit for maximum life time of sensor network. A study of the solar prediction system has shown always previous solar energy information of the previous day in different time slots. Solar harvesting system is very dynamic as well as non-linear, because it is fully based on the different weather condition. Moreover, there are so many issues of the solar harvesting, which is effective harvesting rate of the sensor nodes, such as intensity of sunlight, angle of sunlight and location of deployed sensor nodes. The significance of the prediction works based on the different parameter. A day is divided into M non-overlapping prediction intervals and each slot L consists T duration, means M.L.T=24 ~ hours. However, we can identify above triple function that d is day with l interval and referred to the particulars slot i.

Chapter 4: Proposed Work

Profile energy works on the principle that solar energy reception at a given location follows a pattern. The algorithm leverages the idea to look up the closest energy reception pattern match to the current day and uses it to build an accurate prediction. Profile energy discretizes each day into a set of time slots, and comes up with a clever way of combining the energy harvested in the latest slot with the closest energy generation pattern.

To make the system more accurate and robust to outliers, a weighted mean is taken over each day in memory, with the days observing less error to be given more weight according to the formula:

$$w_j = 1 - \frac{MAE(E_{dj}, C)}{\sum_{j=1}^{P} MAE(E_{dj}, C)}$$

It may be noted that a major drawback of this method is that it does not take into account the error encountered during prediction. Our novel Modified Pro-Energy technique combines the existing algorithm with an error moving average to enable the algorithm to self-correct and learn as the day progresses.

Let us define the error observed $\varepsilon_{d,n}$ on day *d* and timeslot *n* as:

$$\varepsilon_{d,n} = E(d, n-1) - \hat{E}(d, n-1)$$

An exponentially weighted moving average with parameter β is employed on the error, to give us the error correction factor, ϵ as:

$$\epsilon = \beta\epsilon + (1-\beta)\varepsilon$$

Finally, this correction factor is added to our prediction to account for the offset between the current day and previous best-match, as given:

$$\hat{E}(d,n) = \alpha E(d,n-1) + (1-\alpha)\theta + \epsilon$$

Where, θ is our weighted profile coefficient. The intuition behind this idea comes from the assumption that energy generation at a given location follows a certain set pattern, and is only offset by a constant change in intensity or obstruction by clouds or shadows. This mentioned offset is captured in the error moving average in ϵ and overcomes both, problems of insufficient historical data as well as offsets on the current prediction day.

In this paper, we have proposed modified Pro-energy prediction technique for solar harvesting system in the EH-WSNs. This method works based on the past day observation to the derive efficient prediction on the future energy intake. The characteristic profile of day is classified into the different ways, such as sunny, cloudy and rainy. However, harvesting rate can associated in any weather conditions. The modified Pro-energy method divides each day into the N equal-length time slots, thereby reduce time and the memory overhead of prediction approach. Besides, it is very often technique in prediction to divided equally distributed times slots. It is more beneficial to update future harvesting energy estimation only at the pre-defined instants at the particular time.

Profile energy works on the principle that solar energy reception at a given location follows a pattern. The algorithm leverages the idea to look up the closest energy reception pattern match to the current day and uses it to build an accurate prediction. Profile energy discretizes each day into a set of time slots, and comes up with a clever way of combining the energy harvested in the latest slot with the closest energy generation pattern. The modified Pro-energy works same as Pro-energy method only difference to identify prediction error in the previous time slots and also maintained on matrix E(i, j) to keep previous information of the D days. Here, j is a sample within i^{th} day of the previous time slots. The value of α weight factor is similar to EWMA, WCMA and Pro-energy. However, the combination of previous harvested energy H with θ most similar day and introduced prediction error of the previous time slots.

Chapter 5: Result Analysis

Section 5.1: Dataset

The dataset that was used came from NREL Solar Radiation Research Laboratory (BMS). Following is a description of all the data fields available from different instruments from the aforementioned lab.

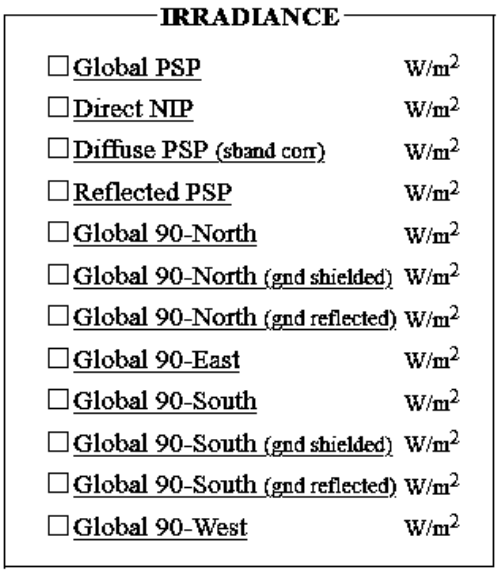

Fig. 1. 12 instrument datasets from NREL (BMS)

A subset of the dataset is shown below:

	DATE (MM/DD /YYYY)	MST	Global CMP22 (vent/cor) [W/m^2]	Global CMP22-1 (cor) [W/m^2]	Global CMP22-2 (cor) [W/m^2]	Global CMP11 (cor) [W/m^2]	Global PSP (vent/cor) [W/m^2]	Global GPP (cor) [W/m^2]	Global Research 1 [W/m^2]	Global UVB-1 [W/m^2]	Global UV-S-B-T [W/m^2]	Global 90-West PSP [W/m^2]	Direct CUVB2 [W/m^2]
0	1/26/2019	07:16	3.97621	4.44060	4.27966	4.83216	8.42773	4.71733	5.19894	-0.001	0.000	1.98487	0.000
1	1/26/2019	07:17	4.96652	5.46627	5.29066	5.97006	9.57426	5.95183	6.60461	-0.001	0.000	2.64293	0.000
2	1/26/2019	07:18	5.75069	6.28833	6.13483	6.76291	10.39120	6.70633	7.65581	-0.001	0.000	2.99068	0.000
3	1/26/2019	07:19	5.33213	5.86996	5.74418	6.39497	10.08150	6.28613	6.44981	-0.001	0.000	3.01208	0.000
4	1/26/2019	07:20	5.08743	5.63157	5.47730	6.20357	10.13100	6.13155	5.68789	-0.001	0.001	3.33309	-0.001
5	1/26/2019	07:21	5.52233	6.09820	5.91677	6.68798	10.69720	6.60658	6.30313	-0.001	0.001	3.83597	0.000
6	1/26/2019	07:22	6.23630	6.84322	6.67704	7.41612	11.46370	7.32754	7.47250	0.000	0.001	4.26931	0.000
7	1/26/2019	07:23	7.13014	7.77046	7.60138	8.30298	12.33090	8.21539	8.73559	0.000	0.002	4.72939	-0.001
8	1/26/2019	07:24	8.33020	8.93881	8.80785	9.29760	13.31010	8.95392	9.37937	0.000	0.002	5.46233	-0.001
9	1/26/2019	07:25	10.18070	10.73690	10.50250	10.41910	14.83740	9.94323	11.63660	0.000	0.002	7.24920	0.000
10	1/26/2019	07:26	13.68170	14.13010	13.63650	11.72600	17.14640	11.12610	14.14230	0.000	0.002	10.53940	0.000
11	1/26/2019	07:27	16.43840	16.67980	16.04660	12.55810	18.87470	11.96700	16.55840	0.000	0.003	12.83990	-0.001
12	1/26/2019	07:28	14.25870	14.59730	14.23350	12.92810	18.00440	12.47980	15.05500	0.001	0.003	10.26120	0.000
13	1/26/2019	07:29	13.41660	13.89800	13.69810	13.85820	18.09850	13.55450	15.62540	0.001	0.004	8.83810	0.000
14	1/26/2019	07:30	15.02990	15.53380	15.33630	15.19730	19.50140	14.90770	17.38150	0.001	0.004	9.99368	0.000
15	1/26/2019	07:31	16.12210	16.68650	16.42640	16.40060	20.80310	16.18890	17.84190	0.001	0.004	10.77480	0.000
16	1/26/2019	07:32	16.93940	17.53810	17.33180	17.51270	21.81660	17.32260	18.86060	0.002	0.005	11.07430	0.000
17	1/26/2019	07:33	17.90580	18.53060	18.29950	18.58580	22.94000	18.44710	20.11550	0.002	0.005	11.62530	0.000
18	1/26/2019	07:34	19.00900	19.60430	19.42700	19.80080	24.05680	19.51130	21.90410	0.002	0.006	12.03720	0.000
19	1/26/2019	07:35	20.49350	21.07120	20.95160	21.46010	25.51200	20.99390	23.70500	0.003	0.006	12.82900	0.000

Fig. 2. Data sample

Section 4.2: Evaluation of the Proposed Algorithm

The proposed algorithm was pitted against EWMA, WCMA and Pro-Energy to establish merit and compare results. Here are some of the key findings.

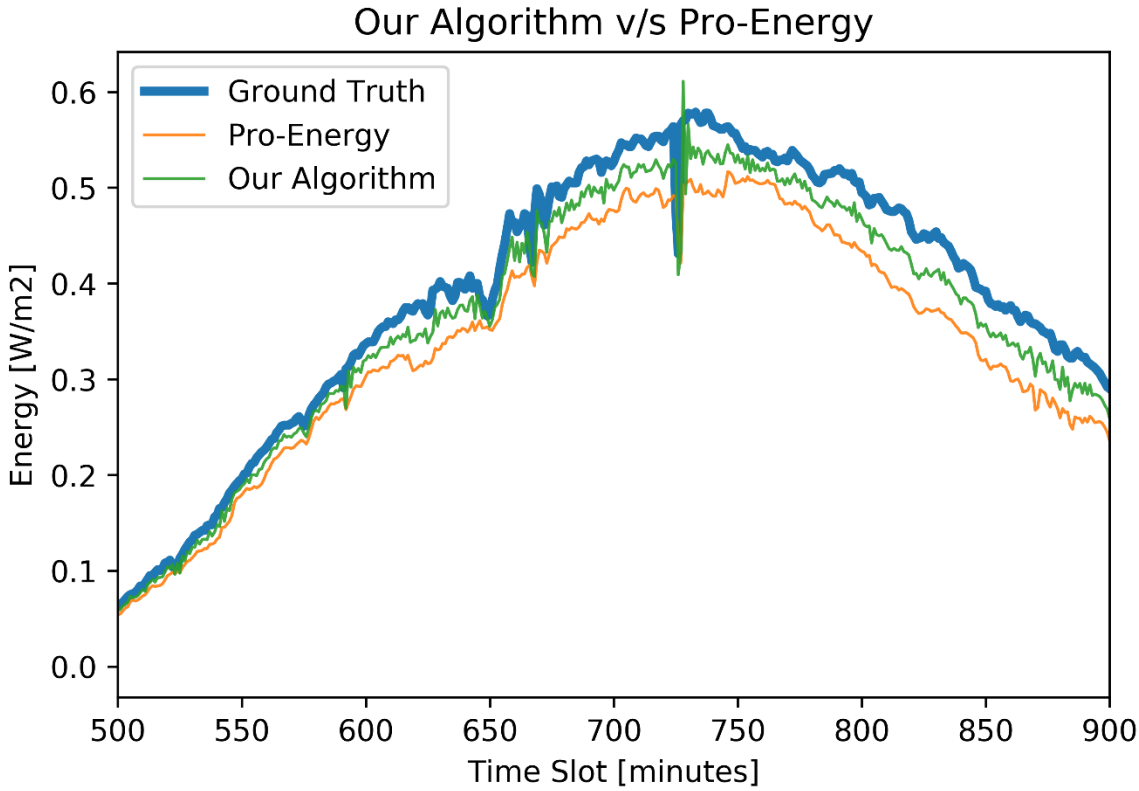

Fig. 3. Pro-Energy vs Proposed Algorithm on Day 10

As can be seen from the figure, our proposed algorithm consistently performs better than the previous state of the art. This results in a much smoother curve with less prediction difference. It is trivial to verify that our intuition for the same (the addition of errors generated during previous mistakes) assists the prediction signal to perform better and better as the day progresses.

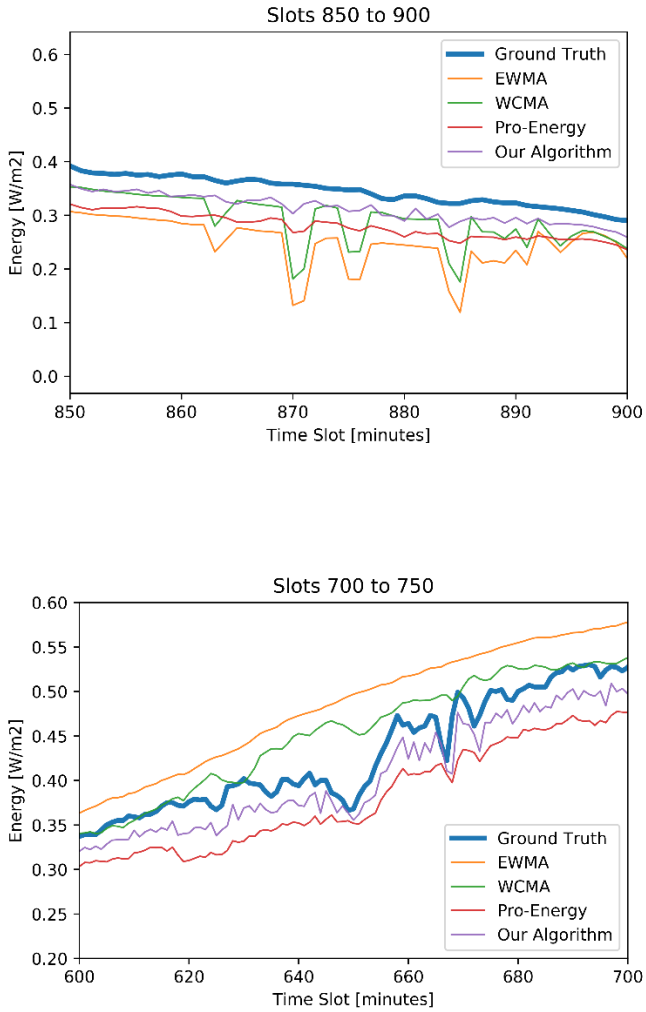

Fig. 4. Comparison of algorithms for smooth (above) and highly varying (below) data points

Here we compare our algorithm against the Exponentially Weighted Moving Average, Weather Condition Moving Average and Profile Energy. As expected, WCMA outperforms EWMA by taking into account the offsets observed by the weather changes in the present day. Pro-Energy outperforms the two moving averages in both situations of high variance (bottom) and low variance (top). It is also easy to note that the proposed method outperforms Pro-Energy as well, resulting in a much closer approximation of the Ground Truth signal. This fact is reflected in the error metric as well.

Section 4.3 Error Metric

The error metric used for evaluation is the Prediction Error Ratio, defined as:

$$Error = \frac{1}{N}\sum_{i=1}^{N}\left|1 - \frac{E}{P}\right|$$

Where, E is the actual and P is our prediction, averaged over all data points.

Section 4.4 Error Analysis

We select four random days from the year and look at the PER values. The trends follow our expectation and Modified Profile Energy outperforms all other methods by a significant margin.

EWMA has no mechanism to check the conditions in the current day, so that it produces high errors in temporary changes. Pro-Energy has a better performance than EWMA as it estimates the energy based on the latest energy generation and the days with the most similar energy generation profile. Finding the most similar profile is performed to compare the total prediction errors among the previous days. Therefore, although the most similar day have the lowest total prediction error, a number of slots may experience significantly wrong energy profile.

	EWMA	WCMA	PRO	MPE
Day 10	1.254554	1.093933	0.933787	0.884650
Day 90	1.837535	1.787515	2.088974	1.069490
Day 200	1.106148	0.983891	0.975873	0.714068
Day 300	3.020421	1.364042	0.931500	0.917892

Table 1. Comparison of EWMA, WCMA, Pro-Energy and Proposed Algorithm on four random different weather days of the year

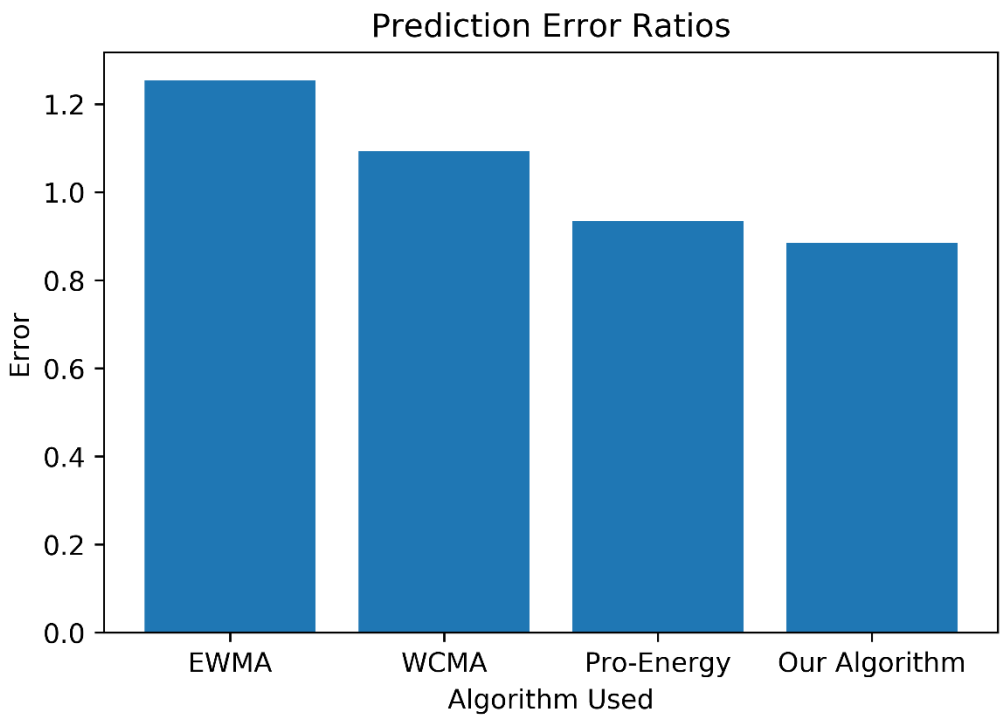

Fig. 5. Average Prediction Error Ratios (bar representation)

Algorithm	PER
EWMA	1.2545536136392341
WCMA	1.093933399644929
Pro-Energy	0.9337872464759861
Mod-Pro-Energy	0.8846498481024798

Table 2. Average Prediction Error Ratios

On running the experiment on a dataset of 365 days and averaging over all days, we notice that Modified Profile Energy outperforms Profile Energy by **5.262%** and outperforms Weather Conditioned Moving Average by **19.13%**, which is a significant gain in accuracy with little computation overhead.

This gain in accuracy helps optimize the sensor node better and helps prolong battery life.

Chapter 5: Conclusion & Future Scope of Work

The application of sensor network has increased in the recent time for different places. One of the major issues is that energy saving, just an example replacement of battery sensor node is too much difficult job in the WSNs. The researcher has been solved the limitation of energy resource using energy harvesting technique. In this paper, we have proposed solar energy harvesting prediction in the EH-WSNs. The energy prediction is initial technique to manage ambient resource. Prediction approach is fully based on the weather condition, because what amount energy will come in the next time slots of the day. This type of justification does not exact result always, but using prediction technique, we can estimation based on of previous day energy information in the respective time slots. Related to prediction issue, modified Pro-energy technique is giving better results and beats other method, such as EWMA, WCMA and Pro-energy. The accuracy of this method is 5.262% better than Pro-energy, its closest counterpart. However, this gain in accuracy helps optimize the sensor node better and helps prolong battery life in the WSNs.

Further, this algorithm may be combined with an efficient scheduling algorithm for packet communication in WSNs to yield better efficiency and reduce the need to replace batteries or run out of juice in the middle of a working cycle.

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